Regime-Switching Asset Allocation via a POMDP

Type: New Application

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The problem: regimes break static allocation

The 60/40 portfolio lost ∼17% in 2022.

A static allocation assumes stable joint return moments. Markets violate this in two ways:

- **Conditional heteroskedasticity:** long quiet periods broken by short crises.
- **Stress correlation:** stocks and bonds sell off together when it hurts most.

Question. Can a decision-theoretic regime overlay, a POMDP solved end-to-end on macroeconomic observables, beat a static benchmark after realistic frictions?

Why POMDP, not a tree, ID, or MDP

Naive choice	Why it fails here
Decision tree	Tree explodes with horizon; no shared state.
Influence diagram	Same blow-up; not natural for repeated rebalance.
Fully-observable MDP	Requires knowing the regime, but it is *latent*.
POMDP	Hidden state, Bayesian belief, QMDP for tractable control.

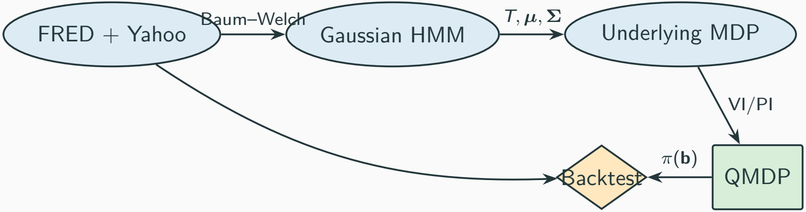
Tools assembled: Bayesian forward filter; Markov chain over regimes; infinite-horizon Bellman recursion solved by value iteration and exact policy iteration; structural results for monotone policies.

POMDP tuple, written once and reused

$$
\begin{aligned}
\mathcal{T} \; & = \{0, 1, 2, \ldots\}, \text{ monthly epochs.} \\
\mathcal{S} \; & = \{s_1, \ldots, s_K\} \text{ latent regimes } (K = 2, 3, 4). \\
\mathcal{A} \; & = \{(0/100), (20/80), \ldots, (100/0)\} \text{ discrete portfolio grid.} \\
T(s' \mid s) \; & \text{ HMM transition matrix; action-independent (regimes are exogenous).} \\
O(\mathbf{o} \mid s) \; & \text{ Gaussian emission on } \mathbf{o} = (\mathrm{VIX}, \mathrm{term\ spread}). \\
R(s, \mathbf{a}) \; & = \mathbb{E}[\, U(\mathbf{a}^\top \mathbf{r}) \mid s] - c\|\mathbf{a} - \mathbf{a}_{\mathrm{prev}}\|_1, \text{ CRRA utility.} \\
\lambda = 0.95 \; & \text{ monthly discount.}
\end{aligned}
$$

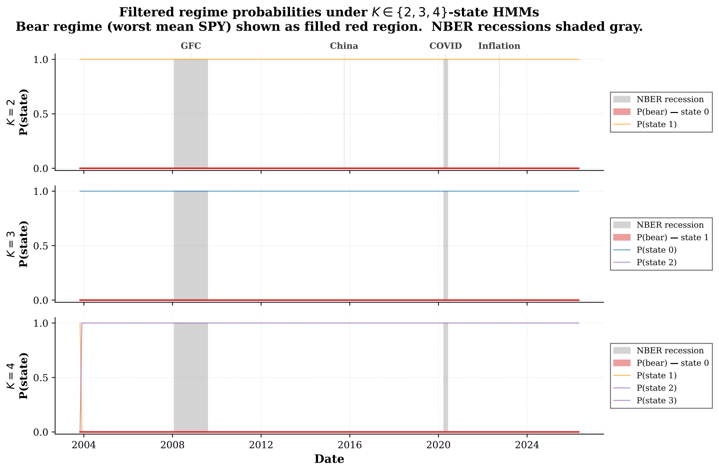
Belief update (Bayesian filter): $\mathbf{b}_t(s') \propto O(\mathbf{o}_t \mid s') \sum_s T(s' \mid s)\, \mathbf{b}_{t-1}(s)$.

Pipeline



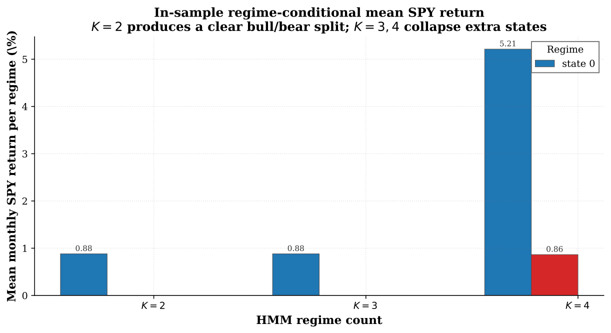
- Raw observations $\rightarrow$ HMM emissions and transitions.
- HMM transition matrix becomes the MDP's T. Reward built by MC.
- VI and PI cross-checked. QMDP folds belief into Q^*.
- Backtest is a walk-forward filter+act loop on real history.

HMM regime timeline: bear bands match crisis dates



Bear regime (filled red) tracks GFC 2008–12, China/oil 2015–16, COVID 2020, inflation 2022. NBER bars (gray) sit inside the recovered bands.

Regime-conditional returns: a clean bull/bear split



Regime	Mean SPY (%/mo)	SD SPY (%/mo)	Mean VIX
Bull (state 0, 87 mo)	+1.16	2.46	15.2
Bear (state 1, 48 mo)	−0.14	6.03	27.7

Underlying MDP: VI and PI agree

The fully-observable MDP satisfies

$$V^*(s) = \max_{\mathbf{a}\in\mathcal{A}}\Big\{R(s,\mathbf{a}) + \lambda\sum_{s'\in\mathcal{S}} T(s' \mid s)\, V^*(s')\Big\}.$$

Value iteration:
$V^{(n+1)} = LV^{(n)}$, stop at
$\|V^{(n+1)} - V^{(n)}\|_\infty < \frac{\varepsilon(1-\lambda)}{2\lambda}$.
125 iters for $\lambda = 0.95$, $\varepsilon = 10^{-4}$.

Policy iteration:
solve $(I - \lambda P_\pi)V = r_\pi$, then
greedy improvement.
2 iters to converge.

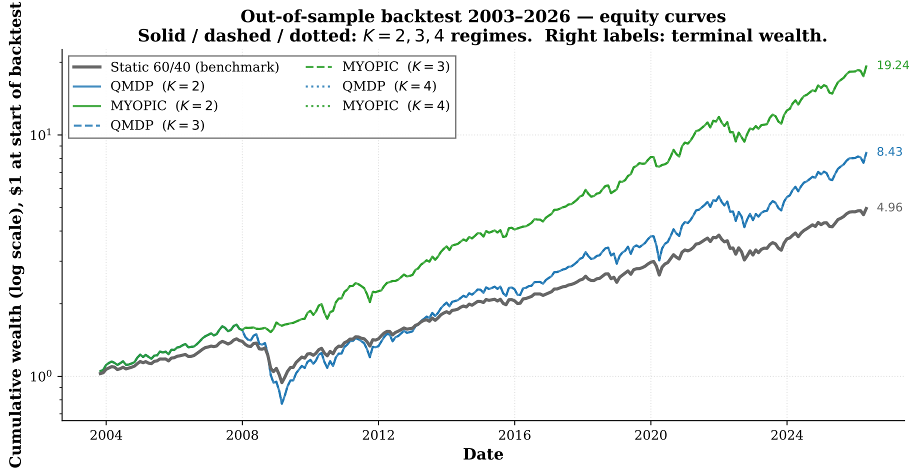
Proposition. VI greedy policy = PI greedy policy, values agree within 2.6×10^{-6}.

QMDP: project belief onto Q^*

$$\pi_{\text{QMDP}}(\mathbf{b}) = \arg\max_{\mathbf{a}\in\mathcal{A}} \sum_{s\in\mathcal{S}} \mathbf{b}(s)\, Q^*(s,\mathbf{a}).$$

- Exact in the fully-observable limit ($\mathbf{b}$ concentrated).
- Treats POMDP as belief-weighted average of MDPs.
- Avoids the intractable belief-space Bellman recursion.
- Implementation: `src/models/qmdp.py::qmdp_action`, single $O(K)$ inner product.

Backtest 2003–2026: equity curves



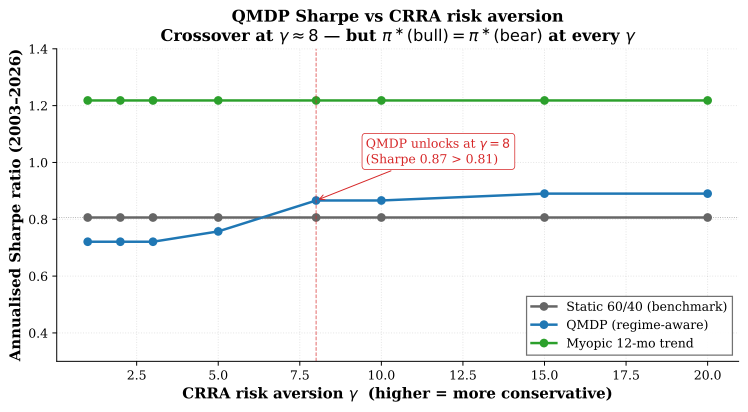
Solid/dashed/dotted: $K = 2, 3, 4$ HMM regime counts. QMDP curves overlap regardless of K; myopic (green) dominates by tracking recent first-moment shifts.

Backtest 2003–2026, headline metrics

Policy	CAGR	Vol	Sharpe	Max DD	Calmar
Static 60/40	7.35%	9.4%	0.81	−34.2%	0.22
QMDP ($\gamma = 2$)	9.90%	14.7%	0.72	−53.0%	0.19
Myopic 12-mo	13.99%	11.3%	**1.22**	−21.1%	**0.66**

Finding. QMDP under log utility *collapses to 100 % stocks*. Risk-tolerance, not regime detection, drives the result. The myopic 12-month trend follower wins on every risk-adjusted measure.

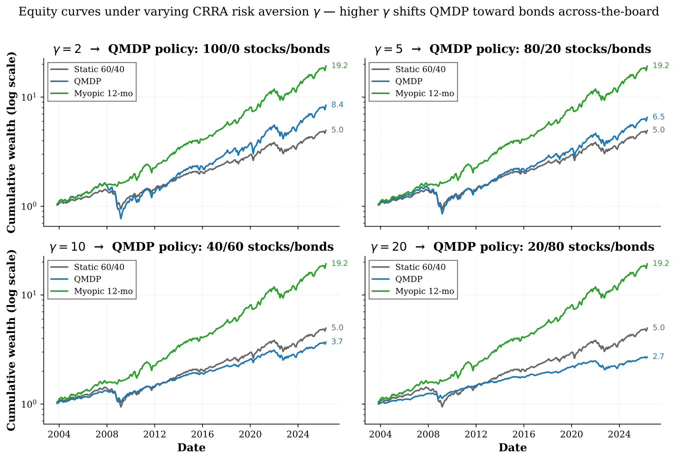
Sensitivity to CRRA γ: QMDP unlocks at $\gamma \geq 8$



Static/myopic are flat in γ by construction. QMDP crosses static at $\gamma \approx 8$.

Caveat: $\pi^*(\text{bull}) = \pi^*(\text{bear})$ at every γ tested.

Effect of γ on equity curves



$\gamma = 2$: 100/0 stocks/bonds. $\gamma = 5$: 80/20. $\gamma = 10$: 40/60. $\gamma = 20$: 20/80. Policy shifts bondward across-the-board; never differentiates regimes.

Conclusions & extensions

What we showed.

- Pipeline correct: VI+PI agree, HMM recovers 2008/2020/2022 stress.
- Under CRRA $\gamma \geq 8$, QMDP Sharpe (0.87–0.89) beats static (0.81).
- But the regime *signal* is detectable while the regime *action* is not, given two macro observables.

Extensions (in priority order).

1. Richer observations: restore HY OAS via FRED API key.
2. Sharpe / CVaR reward shaping: explicitly penalise bear-regime variance.
3. Point-based VI: exploit information value on a 1-D belief simplex (cheap).

Code, data and pipeline: `github.com/takakhoo/ENGS177_Final_Project`